

## ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

## Chapter 8: Precipitation Physics & Freezing Rain (FZRA)

Syllabus Source: CAE Oxford (pp. 249-278) | Pass Target:  $\geq 90\%$  AviationExam

### 1. MECHANISMS OF PRECIPITATION GROWTH

Cloud droplets are microscopic ( $\sim 0.02$  mm) and held aloft by updrafts. Precipitation requires droplet growth by orders of magnitude ( $\sim 1$  to 5 mm):

- Collision-Coalescence Process: Dominant in 'warm clouds' (temperatures  $> 0^\circ\text{C}$ , tropics/maritime). Larger droplets fall faster, colliding with smaller droplets and sweeping them up to form raindrops.
- The Bergeron-Findeisen Process: Dominant in 'cold clouds' (temperatures  $-10^\circ\text{C}$  to  $-40^\circ\text{C}$ ) containing both supercooled water droplets and ice crystals. The saturation vapor pressure over ice is LOWER than over water. Vapor diffuses from water droplets onto ice crystals, causing rapid crystal growth into snowflakes!

### 2. FORMS OF PRECIPITATION

- Drizzle (DZ): Very small water droplets (diameter  $< 0.5$  mm) falling close together. Produced EXCLUSIVELY by Stratus (St) clouds!
- Rain (RA): Liquid water droplets (diameter 0.5 to 5.0 mm). Continuous rain from Nimbostratus (Ns); Intermittent showers from Cumuliform clouds.
- Sleet / Ice Pellets (PL): Translucent frozen raindrops formed when rain falls through a deep freezing layer near the ground.
- Hail (GR): Balls of concentric ice (diameter  $\geq 5$  mm). Produced EXCLUSIVELY by Cumulonimbus (CB) with violent updrafts!

### 3. FREEZING RAIN (FZRA) - THE EXTREME HAZARD

Freezing rain is one of the most hazardous meteorological phenomena in aviation:

- How FZRA Forms: Snow falls from high clouds into a WARM LAYER ALOFT with temperatures  $> 0^\circ\text{C}$ , melting completely into rain. The rain then falls into a SUB-ZERO SURFACE COLD AIR LAYER (temperature  $< 0^\circ\text{C}$ ). The drops become supercooled liquid.
- Impact on Aircraft: Upon impacting the cold airframe, the supercooled rain spreads back over the wing and freezes instantly into CLEAR ICE (Glaze). It can accumulate at over 5 cm per hour, covering aerofoils, freezing flight controls, and causing immediate aerodynamic stall!

### STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

#### [+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Freezing Rain (FZRA) Mechanics

Formation Mechanics: Step 1: Snow falls from high cold layer. Step 2: Falls through a warm inversion aloft with temperature  $> 0^\circ\text{C}$ , melting completely into liquid rain. Step 3: Falls into shallow surface cold layer with temperature  $< 0^\circ\text{C}$ , becoming supercooled droplets. Step 4: On impact with cold airframe, instantly freezes into heavy, transparent CLEAR ICE! Mandatory Cockpit Action: - NEVER DESCEND! Descending keeps you in sub-freezing air with severe icing. - TURN 180 degrees immediately OR CLIMB into the warm inversion layer above!

#### 4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

**Q1: In the Bergeron-Findeisen precipitation process, why do ice crystals grow rapidly while supercooled water droplets do not?**

- [A]: Ice is warmer than water
- [B]: The saturation vapor pressure over ICE is LOWER than the saturation vapor pressure over WATER at the same sub-zero temperature
- [C]: Updrafts only lift ice
- [D]: Sun rays melt water

**[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation**

CORRECT: [B]. Because saturation vapor pressure is lower over ice, a vapor pressure gradient exists from water to ice. Water droplets evaporate and deposit directly onto ice crystals.

**Q2: Under which vertical temperature profile does FREEZING RAIN (FZRA) form?**

- [A]: Temperature permanently below -20°C from surface to cruise
- [B]: A WARM LAYER ALOFT with temperature > 0°C situated above a cold freezing surface layer with temperature < 0°C
- [C]: Standard ISA lapse rate
- [D]: Surface heating

**[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation**

CORRECT: [B]. Falling snow melts into liquid rain in the warm air wedge aloft, then becomes supercooled as it falls through the sub-zero layer near the ground, freezing on contact.

**Q3: Which cloud type is EXCLUSIVELY responsible for producing DRIZZLE?**

- [A]: Cirrus
- [B]: STRATUS (St)
- [C]: Cumulonimbus
- [D]: Altocumulus

**[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation**

CORRECT: [B]. Drizzle consists of uniform droplets < 0.5 mm in diameter, produced exclusively by low-level, stable Stratus clouds.